

Henriette Romero

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EDUCATION

University of Cambridge | *Master of Science in Computer Science* 2015 - 2016

- Achieved First Class distinction with thesis on distributed systems optimization, published in proceedings of Eurosys 2017
- Led university competitive programming team to regional top-3 finish, mentoring 12 junior developers

UC Berkeley | *Bachelor of Science in Electrical Engineering and Computer Science* 2011 - 2015

- Graduated summa cum laude with 3.95 GPA and Regents' and Chancellor's Scholarship recipient
- Published undergraduate research on parallel computing in IEEE Transactions on Computers

EXPERIENCE

Stripe | *Senior Software Engineer, Infrastructure* 2021 - Present

- Architected and deployed a distributed rate-limiting service handling 50M+ requests/day, reducing API timeout errors by 67% and improving P99 latency from 850ms to 120ms
- Led cross-functional team of 5 engineers to migrate legacy monolithic payment processing to microservices, increasing system throughput by 3.2x and enabling 15+ new payment methods
- Mentored 8 junior and mid-level engineers, with 3 promoted to senior roles within 18 months through structured technical growth programs

Datadog | *Senior Software Engineer, Observability Platform* 2019 - 2021

- Designed and implemented real-time anomaly detection engine using time-series analysis, adopted by 2,500+ enterprise customers and generating \$4.2M ARR
- Optimized database query performance across 200+ monitoring dashboards, reducing average load time by 71% (from 2.3s to 650ms) affecting 50K+ active users
- Established automated testing framework and CI/CD pipeline improvements, decreasing deployment failure rate from 12% to 0.8% and release cycle time by 55%

Palantir Technologies | *Senior Software Engineer, Data Platform* 2017 - 2019

- Built high-performance data ingestion pipeline processing 2TB+ daily data volumes, reducing ETL processing time by 64% through algorithmic optimization and caching strategies
- Developed SDKs for 3 major programming languages (Python, Java, Go) enabling 45+ client integrations and adoption across 12 Fortune 500 companies
- Spearheaded code quality initiative implementing static analysis tools, increasing test coverage from 58% to 89% and reducing production bugs by 42%

Figma | *Software Engineer, Multiplayer & Sync* 2016 - 2017

- Enhanced real-time synchronization engine reducing collaboration latency by 73%, enabling seamless teamwork for 100K+ concurrent users during peak hours
- Contributed to WebSocket architecture improvements decreasing memory consumption per connection by 45% and supporting 40% increase in concurrent user limit
- Shipped critical bug fixes and performance optimizations that improved application stability score from 96.2% to 99.1% uptime

PROJECTS

VectorCache | *Rust, C++, Redis, Protocol Buffers* 2022 - Present

- Developed open-source distributed vector database achieving 8.5x faster retrieval times than baseline implementations, reaching 2,500+ GitHub stars in 8 months
- Engineered novel approximate nearest neighbor search algorithm with 94.3% accuracy at 50us query latency, adopted by 40+ enterprise data teams

SyntaxFlow | *Go, TypeScript, GraphQL, PostgreSQL* 2020 - 2022

- Built static analysis tool for detecting code vulnerabilities, scanning 12M+ lines of code across 500+ open-source repositories and identifying 340+ zero-day CVEs
- Achieved 96.8% false-positive rate improvement compared to existing tools through machine learning-based classification, adopted by 85 organizations

MetricsAggregator | *Python, Kafka, ClickHouse, FastAPI* 2019 - 2020

- Created production-grade time-series metrics aggregation service processing 200K+ metrics/sec with <100ms ingestion latency, deployed to 3 companies
- Optimized compression algorithms achieving 31% reduction in storage costs while maintaining query performance within SLA requirements

EXTRACURRICULAR

- Women in Engineering Leadership (WEL) UK Chapter | Senior Mentor & Technical Speaker2020 - Present
- Mentored 18 junior female engineers with 78% advancing to mid-level or leadership positions within 2 years of program participation
 - Delivered keynote on 'Scaling Systems and Teams' at 3 major tech conferences, reaching 1,200+ attendees annually
- HackerOne Bug Bounty & Open Source Community | Active Contributor & Security Researcher2018 - Present
- Discovered and responsibly disclosed 12 critical security vulnerabilities earning \$28,500 in bounties and CVE credits
 - Contributed 45+ pull requests to major open-source projects (CNCF, Linux Foundation) with codebase impact affecting 500M+ developers

TECHNICAL SKILLS

Languages – Go, Python, Rust, TypeScript, Java, C++, SQL

Technologies – Kubernetes, Microservices, Distributed Systems, Real-time Processing, gRPC, Protocol Buffers, GraphQL, WebSockets, Machine Learning, Event Streaming

Tools – Docker, Kafka, Redis, PostgreSQL, ClickHouse, Datadog, Prometheus, Git, GitHub Actions, AWS (EC2, S3, Lambda, RDS), GCP (BigQuery, Compute Engine)